# Aditya Tiwari

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## Experience

**JTP (Japan Third Party Co., Ltd.)** Mar 2026 – Aug 2026
*System Engineer Trainee* *Remote from India*

- Owned end-to-end delivery of a containerized movie recommendation service (React, FastAPI, PostgreSQL), from requirement analysis and design documents through implementation, testing and a formal technical review with the Japan-based team.
- Designed **5 REST endpoints** (search, similar titles, preference matching, random pick) over a ∼10,000-movie catalog, with TF-IDF and cosine-similarity ranking computed per query instead of holding an ∼**800 MB** precomputed matrix in memory.
- Cut poster loading time by fetching TMDB images in parallel on a **10-worker thread pool** with caching, and shipped the stack as 3 Docker Compose services with database health checks, run with a single command.
- Followed Japanese quality practices (PDCA, Kaizen, Horenso, 5S), reporting daily and folding review feedback back into the design, code and installation, usage and technical documentation.

**PwC (Salesforce Launchpad Program)** Jan 2025 – Jul 2025
*Software Engineering Trainee* *Remote, India*

- Shipped **3 enterprise-grade applications** in a 6-month SWE program using OOP design patterns, modular code and database optimization across 4 certified tracks (Java, SQL, RDBMS, Salesforce CRM).
- Designed normalized SQL schemas across 15+ tables and tuned queries with indexing, cutting avg. query time by ∼**35%**.
- Delivered features across 6 Agile/Scrum sprints with peer code reviews, and configured 20+ Salesforce CRM automation rules.

**Altair** Apr 2025 – Jun 2025
*Data Science Intern · Grade: Excellent* *Remote, India*

- Built ML classification and clustering pipelines in RapidMiner over 100K+ records, using cross-validation and hyperparameter tuning to reach **92% model accuracy**.
- Automated preprocessing (correlation, PCA, outlier removal), cutting feature dimensionality by ∼30% and experiment cycles by ∼25%.

## Technical Skills

**Languages**: Python, Java, JavaScript, TypeScript, SQL, HTML5, CSS3
**Web & Backend**: Spring Boot, Spring Data JPA, Hibernate, Spring Security, FastAPI, React JS, REST APIs, Kafka, Microservices
**Databases**: MySQL, PostgreSQL, MongoDB, Redis, SQLAlchemy, Flyway, CTEs, Window Functions, Query Optimization
**Testing & DevOps**: JUnit 5, Mockito, Selenium, Playwright, pytest, Locust, Regression Testing, CI/CD (GitHub Actions, Jenkins)
**Cloud & Tools**: AWS (EC2, S3, RDS), Docker, Kubernetes, Prometheus, Grafana, Splunk, OpenTelemetry, Linux, Bash, Git, GitHub
**AI & ML**: PyTorch, TensorFlow, Scikit-Learn, OpenCV, NLP, Hugging Face, LangChain, OpenAI API
**Core CS**: Data Structures & Algorithms, OOP, Design Patterns, Multithreading, System Design, DBMS, OS, Computer Networks
**Practices**: SDLC, Agile/Scrum, Kanban, Code Reviews, Secure Coding, OWASP, PDCA, Troubleshooting

## Education

**GITAM University, Visakhapatnam** 2022 – 2026
*B.Tech in Computer Science Engineering* *CGPA: 7.44*

**Relevant Coursework:** Data Structures, DBMS, Operating Systems, Computer Networks, Software Engineering, Machine Learning

## Projects

**ServScope** | *Distributed Tracing & Observability Platform* 2026 · GitHub

- Built an observability platform that ingests OpenTelemetry traces and logs from a 4-service app, builds per-minute metrics at ingest, and links alerts to the traces showing the root cause, with a React + TypeScript dashboard.
- Designed reliable, idempotent ingestion (data confirmed only after commit, duplicate-safe keys, 503 backpressure), verified by failure drills (database outage, crash mid-write, traffic flood).
- Profiled with py-spy and rewrote the hot database path in SQLAlchemy Core, raising throughput **about 5x (3,400 to 17,000 spans/s** on one core), measured with Locust; CI runs **220 tests**, Playwright and security scans.
- **Tech stack:** Python, FastAPI, PostgreSQL, SQLAlchemy, OpenTelemetry, React, TypeScript, Docker, Locust, GitHub Actions

**ClaimFlow** | *Event-Driven Health Claims Processing* 2026 · GitHub

- Built two Spring Boot microservices that process health-insurance claims over Apache Kafka, checking coverage, covered services, deductible, coinsurance and annual maximum to approve, deny or pend each claim.
- Published events only after the database commit, made decision handling idempotent, and added retries with a dead-letter topic; stored claims in MySQL (Flyway migrations) and an audit timeline in MongoDB.
- Covered by **32 JUnit 5 tests** (embedded Kafka, full two-service flow) and **4 Selenium UI tests**; GitHub Actions runs them plus a Docker Compose smoke test, with structured logs shipped to Splunk.
- **Tech stack:** Java, Spring Boot, Apache Kafka, MySQL, MongoDB, JUnit 5, Selenium, Docker, GitHub Actions, Splunk

**AI Med Card** | *Full-Stack Healthcare Platform* 2026 · GitHub · Live Demo

- Built a healthcare platform with Patient, Doctor and Superadmin portals on a FastAPI REST backend and PostgreSQL, with bcrypt + JWT role-based auth, deployed live with Docker on Render.
- Added AI analysis of uploaded PDF and image reports, appointment booking and an anonymized QR emergency card, covered by **19 pytest API tests** and Playwright end-to-end tests.
- **Tech stack:** Python, FastAPI, PostgreSQL, SQLAlchemy, JWT, pytest, Playwright, Docker, Render, JavaScript, HTML5, CSS3

## Certifications

- PwC — Salesforce Launchpad Program; tracks in IT Fundamentals, Java Programming, RDBMS, Salesforce CRM (2025)
- Coursera — Python Data Structures (Univ. of Michigan); C for Everyone: Programming Fundamentals (UC Santa Cruz)
- Coursera — TCP/IP and Advanced Topics; Fundamentals of Network Communication (Univ. of Colorado System)
- Coursera — Introduction to Cloud Computing (IBM); Mastering Generative AI Workshop — IIT Hyderabad (2024)

## Achievements

- Top teams nationally at **Walmart Global Tech Hackathon 2025** (Smart Retailer Buddy), competing with 500+ teams.
- Built and demoed the KrishiBal AI farmer portal at **Smart India Hackathon 2025** (live).
- E-Club Co-Coordinator, GITAM University — led 10+ entrepreneurship events and speaker workshops, engaging 200+ students.